Pranav Purwar

Systems Engineer | JVM Internals & Android Virtualization

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Summary

Systems Engineer specializing in *JVM internals*, *compiler toolchains*, *AI infrastructure*, and *Android OS-level virtualization*. Creator of **Cosmic IDE** (**686+ stars**, **100+ forks**) featuring custom glibc runtime with recursive process wrapping, and **Bryte AI Platform** with *native FSRS v4 implementation* and *GraphRAG retrieval*. Core contributor to open-source platforms with **1.2M+ downloads** and **30k+ daily users**.

Experience

Bryte AI Platform (Open-Source) | Creator & Lead Developer — GitHub
Sept 2024 – present
2 years

- Designed **Kotlin Multiplatform** backend infrastructure for multi-modal document ingestion (PDF/OCR via Mistral SDK, YouTube via open-source extractor, web scraping with recursive HTML cleaning and LLM-powered hierarchical reconstruction).
- Implemented production-grade **GraphRAG** with Neo4j: **1024-dim vector index** with cosine similarity, weighted Cypher queries (configurable docBias/crossDocPenalty), automatic chunk interlinking with semantic thresholds (**≥0.65**), and real-time graph visualization endpoints.
- Engineered native Kotlin port of **FSRS v4** (spaced-repetition algorithm) with exact mathematical fidelity to upstream.
- Architected stateful AI agent orchestration using JetBrains Koog framework with autonomous history compression for **100k+ token context windows**, plus resilient MarkdownStreamingParser for real-time fragmented token processing with heuristic recovery for broken table structures.
- Built full-duplex WebSocket communication via kotlinx.rpc: multiplexed SessionService and FlashcardService over single connections with type-safe contracts, enabling **real-time streaming** of generated content to clients the millisecond they are parsed.

Cosmic IDE | Creator & Architect — GitHub
Aug 2021 – present
5 years 1 month

- Architected full Linux OS environment on Android without root, proot, or containers using custom glibc runtime with **LD_PRELOAD shims** that intercept filesystem, DNS, and process execution calls.
- Designed **recursive process sandboxing**: execve/execvp/posix_spawn interception ensures all child processes inherit shim chain, preventing any process from escaping the sandbox.
- Engineered **native PTY implementation** with proper terminal emulation, foreground process group signal handling, and Ctrl+C propagation for interactive shells.
- Built plugin-based IDE architecture with **11 language plugins** (Rust, Java, Kotlin, Scala, Gradle, Maven, Go, Gleam, C/C++, Python, Lua) using dynamic loading and marketplace system.
- Implemented complete Arch Linux ARM runtime with **pacman integration**, fake root support, and automated glibc environment generation.
- Developed full-featured IDE with **real-time code analysis**, quick fixes, and LSP support across all plugins directly on mobile.

OpenJDK & Kotlin Compiler Ports (Android) — OpenJDK | Kotlin Compiler
Sept 2023 – present
3 years

- Ported javac (**OpenJDK 27 EA**) and kotlinc (**2.4+**) to Android Runtime (**ART**), maintaining patches synced with upstream releases.
- Rewrote javac's internal platform resolution to bypass Android's restricted NIO by implementing **custom ZipFileSystem** for ct.sym, and patched classloading incompatibilities.
- Engineered memory reclamation layer using HiddenApiBypass to invoke Android's internal NioUtils.freeDirectBuffer, preventing **fatal OOM crashes** during memory-mapped JAR parsing.

Official Kotlin Compiler Contribution (JetBrains/kotlin #5146) — Pull Request
Nov 2023

- Contributed to the official Kotlin compiler by resolving a **critical memory leak**, improving performance for fast JAR file system operations.

Sketchware-Pro | Core Team & Technical Lead — GitHub
Aug 2021 – present
5 years 1 month

- Part of core team for the IDE (**1.6k+ stars**, **100k+ MAU**, **58+ contributors**) powered by the **ported Kotlin compiler**.
- Built build-system pipeline supporting Java 9/11 compilation; integrated **R8 Code Shrinker** reducing output **APK sizes by ~40%**.

Reef (Digital Wellbeing & Productivity) [Reef](#) Sept 2024 – present 2 years

- Architected system-level digital wellbeing suite navigating strict Android background execution limits and battery optimization constraints to maintain reliable per-app tracking.
- Solved severe Android **OEM fragmentation** by engineering centralized UsageCalculator normalizing telemetry across diverse manufacturer implementations (MIUI, OneUI, etc.).
- Implemented context-aware notification routing via Accessibility/Notification Listener services with complex time-boundary tracking for overnight routines.

Skills

Languages: Kotlin, Java, Python, TypeScript, C, JVM Bytecode, DEX/ART

Compiler & Toolchain: OpenJDK, Kotlin, ART, LSP, Gradle, R8/D8, Hidden API Bypass, ELF/PTY

Backend & AI Infrastructure: Ktor, Neo4j (GraphRAG/Cypher), GraphRAG, FSRS v4, LLM Agents**, Koog Framework, Mistral/Cohere/Gemini APIs, SSE Streaming

Mobile & Platforms: Android SDK, Jetpack Compose, Kotlin Multiplatform (KMP), Material You, F-Droid

Systems Programming: glibc, Dynamic Linking, LD_PRELOAD, Process Virtualization, Custom Linkers, Recursive Sandboxing